

Highlights

- CPA and eCPA EoS validated for CO₂+H₂O/brine from 0 to 425 °C, 1 to 1500 bar
- Newton–Raphson solvers with analytical Jacobians approach the speed of a cubic EoS
- 100% flash convergence across ~10,000 conditions; AARE 6.5% vs. experiments
- Pre-computed solution table cuts flash cost by over two orders of magnitude
- Speed and robustness demonstrated in a prototype CO₂ reservoir simulator